Category Theory for Mathematical Physics — Exercise Sheet 9

Ribbon categories, Temperley–Lieb and $\mathrm{SU}(2)_k$

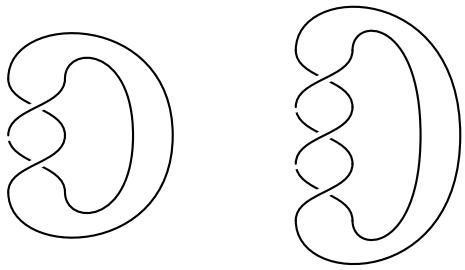
Variant: Solutions

Conventions as in Lecture 9: the crossing is $c = A\,\mathrm{id} + A^{-1}U$ (Equation (9.4)), the loop is $\delta = -A^2 - A^{-2}$ (Equation (9.3)), $q = A^2$ and $[n] = [n]_q = (q^n - q^{-n})/(q - q^{-1})$, so $\delta = -[2]$. In $\mathrm{TL}_n(\delta)$, U_i is the cup-cap on strands $i, i+1$; a product XY means Y first (drawn below). The *closure* of a diagram on n strands joins each top endpoint to the bottom one below it on the right, and is evaluated by Equation (9.4) and Equation (9.3). p_n are the Jones–Wenzl projectors (Theorem 9.7). A coloured strand is labelled by its number n of strands (its colour), as in $(2, p_2)$, θ_n, d_n; only in Exercise 4(ii)–(iii) are the simple objects of $\mathrm{SU}(2)_3$ labelled by spins $j = n/2$, as in Theorem 9.9. $\varphi = (1+\sqrt{5})/2$.

Exercise 1 (Hopf link, trefoil, Jones). (i) Draw the Hopf link as the closure of c^2 and the right-handed trefoil as the closure of c^3, on two strands, each strand oriented upward along the braid (so every crossing is positive). Write c^2 and c^3 as $\alpha + \beta U$ in $\mathrm{TL}_2(\delta)$, and close them up to get $\langle \mathrm{Hopf} \rangle$ and $\langle \mathrm{trefoil} \rangle$.

(ii) With the writhe w (the number of positive crossings minus negative ones), the Jones polynomial is $V_L(t) = (-A^3)^{-w}\langle L \rangle / \langle O \rangle$ at $A = t^{-1/4}$ (Lecture 9). Compute it for the Hopf link and the trefoil.

Solution. (i) The closures, the crossings drawn by Equation (9.4)'s convention:



With $U^2 = \delta U$: $c^2 = A^2 + 2U + A^{-2}\delta U = A^2 + (1 - A^{-4})U$, since $2 + A^{-2}\delta = 1 - A^{-4}$. Then $c^3 = (A + A^{-1}U)(A^2 + (1 - A^{-4})U) = A^3 + \big(A - A^{-3} + A + A^{-1}\delta - A^{-5}\delta\big)U$, and with $A^{-1}\delta = -A - A^{-3}$, $-A^{-5}\delta = A^{-3} + A^{-7}$, the coefficient is $A - A^{-3} + A^{-7}$. The closure of id is δ^2 and that of U is δ, so

$$\langle \mathrm{Hopf} \rangle = A^2\delta^2 + (1 - A^{-4})\delta = \delta(-A^4 - 1 + 1 - A^{-4}) = \delta(-A^4 - A^{-4}),$$
$$\langle \mathrm{trefoil} \rangle = A^3\delta^2 + (A - A^{-3} + A^{-7})\delta = \delta(-A^5 - A^{-3} + A^{-7}),$$

the first as in Equation (9.5).

(ii) Both crossings of the Hopf link and all three of the trefoil are positive: $w = 2$ and $w = 3$. Hopf: $(-A^3)^{-2}(-A^4 - A^{-4}) = -A^{-2} - A^{-10}$, and $A = t^{-1/4}$ gives $-t^{1/2} - t^{5/2}$. Trefoil: $(-A^3)^{-3}(-A^5 - A^{-3} + A^{-7}) = A^{-4} + A^{-12} - A^{-16} = t + t^3 - t^4$. (The left-handed trefoil, the closure of c^{-3}, has $t^{-1} + t^{-3} - t^{-4}$: here the Jones polynomial tells the trefoil from its mirror image, though it does not do so for every knot. Reversing the orientation of one component of the Hopf link makes both its crossings negative, and changes V to $-t^{-1/2} - t^{-5/2}$.)

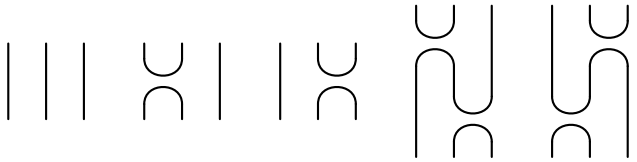
Exercise 2 (TL$_3$ and Reidemeister III). (i) Draw the five diagrams $1, U_1, U_2, U_1U_2, U_2U_1$ on three strands, and show that the product of any two of them is a multiple of one of them. Conclude $\dim \mathrm{TL}_3 = 5$, given that the five are independent (they are, on $V_1^{\otimes 3}$ for generic A).

(ii) With $c_i = A + A^{-1}U_i$, show from the TL relations alone that

$$c_1c_2c_1 = A^3 + A(U_1 + U_2) + A^{-1}(U_1U_2 + U_2U_1) = c_2c_1c_2.$$

(iii) Deduce $c_1c_2c_1^{-1} = c_2^{-1}c_1c_2$, the braid relation with two of its crossings inverted.

Solution. (i) The diagrams, read upward (the product XY puts Y below):



that is 1, U_1, U_2, U_1U_2 and U_2U_1 (in the last two the middle strand snakes from one side to the other). Every word in U_1, U_2 reduces, by $U_i^2 = \delta U_i$ (a closed loop) and $U_1U_2U_1 = U_1$, $U_2U_1U_2 = U_2$ (a snake straightened), to a power of δ times an alternating word of length at most 2, that is one of the five; so the five span TL_3, and being independent they are a basis. (This is the Catalan number $C_3 = 5$: the non-crossing matchings of six points.)

(ii) Expanding,

$$\begin{aligned} c_1c_2c_1 &= A^3 + A(2U_1 + U_2) + A^{-1}(U_1U_2 + U_2U_1 + U_1^2) + A^{-3}U_1U_2U_1 \\ &= A^3 + AU_2 + A^{-1}(U_1U_2 + U_2U_1) + (2A + A^{-1}\delta + A^{-3})U_1, \end{aligned}$$

and $2A + A^{-1}\delta + A^{-3} = 2A - A - A^{-3} + A^{-3} = A$. The result is symmetric under $U_1 \leftrightarrow U_2$, and the same computation with 1 and 2 exchanged gives $c_2c_1c_2$: they are equal. Only the TL relations and $\delta = -A^2 - A^{-2}$ were used; this is Reidemeister III for the bracket.

(iii) The crossings are invertible ($c^{-1} = A^{-1} + AU$, by RII). Multiplying $c_1c_2c_1 = c_2c_1c_2$ by c_2^{-1} on the left and c_1^{-1} on the right gives $c_2^{-1}c_1c_2 = c_1c_2c_1^{-1}$: the relation of (ii) with two crossings inverted, again an invariance of the bracket.

Exercise 3 (The projectors p_2 and p_3). Throughout, A is such that $[2], [3] \neq 0$.

(i) Show that $p_2 = 1 - U/\delta = 1 + U/[2]$ is idempotent with $Up_2 = 0$, and that its closure is $[3]$.

(ii) Build $p_3 = p_2 \otimes 1 + \frac{[2]}{[3]}(p_2 \otimes 1)U_2(p_2 \otimes 1)$ in TL_3 and show that

$$p_3 = 1 + \frac{[2]}{[3]}(U_1 + U_2) + \frac{1}{[3]}(U_1U_2 + U_2U_1).$$

(iii) Check $U_1p_3 = 0$, and compute the closure of p_3.

Solution. (i) $p_2^2 = 1 + 2U/[2] + U^2/[2]^2 = 1 + (2/[2] + \delta/[2]^2)U = 1 + (2/[2] - 1/[2])U = p_2$, using $\delta = -[2]$. $Up_2 = U + U^2/[2] = (1 + \delta/[2])U = 0$. The closure is $\delta^2 + \delta/[2] = \delta^2 - 1 = [2]^2 - 1 = [3]$, as in Equation (9.7).

(ii) With $p_2 \otimes 1 = 1 + U_1/[2]$, $(p_2 \otimes 1)U_2(p_2 \otimes 1) = U_2 + (U_1U_2 + U_2U_1)/[2] + U_1U_2U_1/[2]^2 = U_2 + (U_1U_2 + U_2U_1)/[2] + U_1/[2]^2$. Multiplied by $[2]/[3]$ and added to $1 + U_1/[2]$: the coefficient of U_2 is $[2]/[3]$, that of U_1U_2 and U_2U_1 is $1/[3]$, and that of U_1 is $\frac{1}{[2]} + \frac{1}{[2][3]} = \frac{[3]+1}{[2][3]} = \frac{[2]}{[3]}$, by $[2]^2 = [3] + 1$.

(iii) $U_1p_3 = U_1 + \frac{[2]}{[3]}(\delta U_1 + U_1U_2) + \frac{1}{[3]}(\delta U_1U_2 + U_1)$, using $U_1^2 = \delta U_1$ and $U_1U_2U_1 = U_1$. The coefficient of U_1U_2 is $([2] + \delta)/[3] = 0$, and that of U_1 is $1 + (\delta[2] + 1)/[3] = ([3] - [2]^2 +$

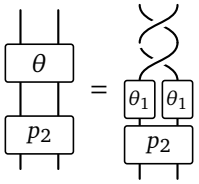
$1)/[3] = 0$. By the symmetry $1 \leftrightarrow 2$, $U_2 p_3 = 0$ as well. The closure is $\delta^3 + 2\frac{[2]}{[3]}\delta^2 + \frac{2}{[3]}\delta = -[2]^3 + \frac{2[2]^3 - 2[2]}{[3]} = -[2]\big([3] - 1\big) = -\big([2]^3 - 2[2]\big) = -[4]$, using $[2]^2 - 1 = [3]$; this is $(-1)^3[4]$, as Theorem 9.7 says.

Exercise 4 (Coloured twists, and Fibonacci inside $\mathrm{SU}(2)_3$). (i) Draw the twist axiom of Definition 9.1 for $V_1 \otimes V_1$ (Equation (9.1)) with p_2 inserted at the bottom, and use it, with $\theta_1 = -A^3$, to show that the coloured strand $(2, p_2)$ has twist A^8, as Equation (9.10) says.

(ii) At $k = 3$, $A = e^{i\pi/10}$: list d_j and θ_j for $j = 0, \frac{1}{2}, 1, \frac{3}{2}$ after the pivotal flip of Theorem 9.9. Show that $\{0, 1\}$ is Fibonacci, with its d_τ and θ_τ, and identify $\frac{3}{2}$.

(iii) Check that $\theta_{1/2} = \theta_1 \theta_{3/2}$.

Solution. (i) On $V_1 \otimes V_1$ the twist axiom reads $\theta_{V_1 \otimes V_1} = c^2(\theta_1 \otimes \theta_1) = A^6 c^2$, and the twist is natural, so it commutes with p_2 and acts on the image of p_2 by restriction. There $c\, p_2 = (A + A^{-1}U)p_2 = Ap_2$, so $c^2 p_2 = A^2 p_2$ and the twist of $(2, p_2)$ is $A^2 \cdot A^6 = A^8 = (-1)^2 A^{2\cdot 4}$. In pictures: the twist of the doubled strand above p_2 is a twist on each strand ($(-A^3)^2$) and one full twist of the pair, two crossings (A^2 on p_2):



(ii) At $k = 3$, $q = e^{i\pi/5}$ and $[n] = \sin(n\pi/5)/\sin(\pi/5)$. After the flip, $d_j = [2j + 1]$: $d_0 = 1$, $d_{1/2} = [2] = 2\cos(\pi/5) = \varphi$, $d_1 = [3] = \varphi$, $d_{3/2} = [4] = 1$; and $\theta_j = e^{2\pi i\, j(j+1)/5}$: $\theta_0 = 1$, $\theta_{1/2} = e^{3\pi i/10}$, $\theta_1 = e^{4\pi i/5}$, $\theta_{3/2} = e^{3\pi i/2} = -i$. The truncated rule gives $1 \otimes 1 \cong 0 \oplus 1$ (the channel 2 is cut, $2 > k - 2 = 1$), so $0, 1$ fuse like $1, \tau$, with $d_1 = \varphi = d_\tau$ and $\theta_1 = e^{4\pi i/5} = \theta_\tau$ (Lecture 7): Fibonacci, with the course's chirality. The object $\frac{3}{2}$ has $\frac{3}{2} \otimes \frac{3}{2} \cong 0$ and $d = 1$: it is invertible, with spin $-i$, an antisemion (Theorem 7.8).

(iii) $\theta_1 \theta_{3/2} = e^{4\pi i/5} e^{-i\pi/2} = e^{3\pi i/10} = \theta_{1/2}$. Since $\frac{1}{2} \cong 1 \otimes \frac{3}{2}$ (one channel), the ribbon relation $\theta_{1/2} = \theta_1 \theta_{3/2} R^{1,\frac{3}{2}} R^{\frac{3}{2},1}$ shows that 1 and $\frac{3}{2}$ braid trivially around each other (their double braiding is 1), consistent with Sheet 5's $\dim \mathrm{SU}(2)_3 = 2 \dim \mathrm{Fib}$. That $\mathrm{SU}(2)_3$ is then the product of Fibonacci and the antisemion uses the Deligne product of Lecture 12 and is not proved here.

Exercise 5 (The $k = 1$ sign check). Take $k = 1$, $A = e^{i\pi/6}$.

(i) Compute δ and $[3]$, and show that p_2 is negligible (Definition 9.8); which coloured strands survive?

(ii) Compute d_1 and θ_1 for the colour 1 (spin $\frac{1}{2}$) in the Temperley–Lieb pivotal structure and after the flip.

(iii) Compute the number by which c acts on the cup $\mathbb{C} \to V_1 \otimes V_1$, and compare with the semion of Lecture 7.

Solution. (i) $q = e^{i\pi/3}$, so $\delta = -(q + q^{-1}) = -2\cos(\pi/3) = -1$ and $[3] = q^2 + 1 + q^{-2} = 1 + 2\cos(2\pi/3) = 0$. The closure of p_2 is $[3] = 0$. For any g on two strands, $\mathrm{tr}(gp_2) = \mathrm{tr}(gp_2p_2) = \mathrm{tr}(p_2gp_2)$, and p_2gp_2 is a multiple of p_2 (on two strands, $p_2Up_2 = 0$ and $p_21p_2 = p_2$), so $\mathrm{tr}(gp_2) = \lambda\,\mathrm{tr}(p_2) = 0$: p_2 is negligible. The surviving coloured strands are 0 and 1, and $1 \otimes 1 \cong 0$.

(ii) In the TL structure $d_1 = \delta = -1$ and $\theta_1 = -A^3 = -e^{i\pi/2} = -i$. The flip multiplies both by $(-1)^1$: $d_1 = 1$, $\theta_1 = i$, as Theorem 9.9 says with $\theta_{1/2} = e^{2\pi i(3/4)/3} = e^{i\pi/2} = i$.

(iii) $U = v \circ u$ with $u \circ v = \delta$, so $Uv = \delta v$, and $c\,v = (A + A^{-1}\delta)v = (A - A^{-1})v = 2i\sin(\pi/6)\,v = i\,v$: the R-symbol $R_0^{11} = i$. This is the semion of Theorem 7.8, $c_{s,s} = i$, whose spin $\theta_s = c_{s,s} = i$ is the flipped value: the TL structure's $-i$ differs by the sign $\kappa_{1/2} = -1$ of Lecture 4, which the flip removes.

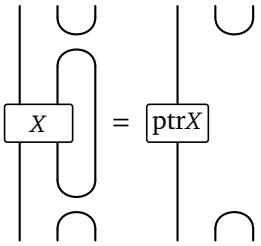
Exercise 6 ⋆. *The Jones–Wenzl recursion in general.* Write $\mathrm{ptr}(X)$ for the closure of the last strand of X on the right, a map from n strands to $n-1$, and suppose $[m] \neq 0$ for $m \leq n+1$. Suppose p_n is an idempotent killed by $U_1, \dots, U_{n-1}$ (on both sides), with $(p_{n-1} \otimes 1)p_n = p_n = p_n(p_{n-1} \otimes 1)$ and $\mathrm{ptr}(p_n) = -\frac{[n+1]}{[n]}\, p_{n-1}$.

(i) Show that $U_n(X \otimes 1)U_n = (\mathrm{ptr}(X) \otimes 1 \otimes 1)\, U_n$ for X on n strands (draw it for $n = 2$), and deduce that $p_{n+1} = p_n \otimes 1 + \frac{[n]}{[n+1]}(p_n \otimes 1)U_n(p_n \otimes 1)$ is killed by U_n.

(ii) Show that p_{n+1} is killed by $U_1, \dots, U_{n-1}$, is idempotent, and is absorbed by $p_n \otimes 1$.

(iii) Show that $\mathrm{ptr}(p_{n+1}) = -\frac{[n+2]}{[n+1]}p_n$, and conclude by induction that the closure of p_n is $(-1)^n[n+1]$.

Solution. (i) Each U_n is a cap (below) followed by a cup (above) on strands $n, n+1$. In $U_n(X \otimes 1)U_n$ the cup of the lower U_n feeds strand n of X and the extra strand $n+1$, and the cap of the upper U_n joins them again above X: the last strand of X is closed on the right, which is $\mathrm{ptr}(X)$ on the first $n-1$ strands, and what remains is the lower cap and the upper cup, that is U_n. For $n = 2$:



With $X = p_n$: $U_n(p_n \otimes 1)U_n(p_n \otimes 1) = -\frac{[n+1]}{[n]}(p_{n-1} \otimes 1 \otimes 1)U_n(p_n \otimes 1)$. Since $p_{n-1} \otimes 1 \otimes 1$ acts on the first $n-1$ strands, it commutes with U_n, and $(p_{n-1} \otimes 1 \otimes 1)(p_n \otimes 1) = p_n \otimes 1$ by absorption. So $U_n p_{n+1} = U_n(p_n \otimes 1) + \frac{[n]}{[n+1]} \cdot \big(-\frac{[n+1]}{[n]}\big)U_n(p_n \otimes 1) = 0$. The mirror computation gives $p_{n+1}U_n = 0$.

(ii) For $i < n$, U_i acts on the first n strands, so $U_i(p_n \otimes 1) = (U_i p_n) \otimes 1 = 0$ kills both terms of p_{n+1}; likewise on the right. Absorption: $(p_n \otimes 1)p_{n+1} = p_{n+1} = p_{n+1}(p_n \otimes 1)$, because both terms of p_{n+1} begin and end with $p_n \otimes 1$, an idempotent. Then $p_{n+1}^2 = p_{n+1}\big(p_n \otimes 1 + \frac{[n]}{[n+1]}(p_n \otimes 1)U_n(p_n \otimes 1)\big) = p_{n+1} + \frac{[n]}{[n+1]}(p_{n+1}U_n)(p_n \otimes 1) = p_{n+1}$.

(iii) Closing the last strand of $p_n \otimes 1$ closes a free strand: δp_n. Closing the last strand of $(p_n \otimes 1)U_n(p_n \otimes 1)$ straightens the cup-cap into a single strand (a snake), leaving $p_n p_n = p_n$. So $\mathrm{ptr}(p_{n+1}) = \big(\delta + \frac{[n]}{[n+1]}\big)p_n = \frac{-[2][n+1]+[n]}{[n+1]}p_n = -\frac{[n+2]}{[n+1]}p_n$, by $[2][n+1] = [n+2] + [n]$ (from $(q+q^{-1})(q^{n+1} - q^{-n-1}) = q^{n+2} - q^{-n-2} + q^n - q^{-n}$). This is the hypothesis for $n+1$, so the induction runs, starting from $p_1 = 1$, p_0 the empty diagram, $\mathrm{ptr}(p_1) = \delta = -\frac{[2]}{[1]}$. The closure of p_n is n partial traces: $\prod_{m=1}^{n}\big(-\frac{[m+1]}{[m]}\big) = (-1)^n[n+1]$, as in Theorem 9.7.